

Estimating the Risk of Depression and Care Burden among Dementia Caregivers: A Feasibility Study

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Abstract—Caregivers of people living with dementia (PwD) are highly susceptible to depression due to the care burden they experience. While many caregivers often neglect their own mental health and do not seek necessary medical services, caregivers often interact with social workers due to their care responsibilities and explain their perceived burden and depressive symptoms to the social workers. A key to directing caregivers' attention to early screening and intervention for depression is the ability to estimate the risk of depression and perceived care burden in these frequent conversational interactions. In this paper, we demonstrate that an in-depth analysis of caregivers' demographic information and their narratives about their care experiences can estimate the risk of depression and burden using Natural Language Processing (NLP) and machine learning algorithms. Specifically, our models were trained to estimate the risk of depression and perceived care burden based on the Patient Health Questionnaire-8 (PHQ-8) and Zarit Burden Interview (ZBI) scores from 65 caregivers of PwD. The resulting models estimated the risk of depression and burden with F1 Scores of 0.72 and 0.76, respectively. Early screening and intervention for depression facilitated by our approach has the potential to significantly improve the well-being of caregivers, ultimately enhancing the quality of care provided to PwD.

Index Terms—Early intervention, mental health, prediction, depression, burden, informal caregivers, dementia, Patient Health Questionnaire, Zarit Burden Interview

I. INTRODUCTION

THE global population is aging, leading to an increase in dementia cases worldwide [1]. Caring for people living with dementia (PwD) is a demanding responsibility, often placing a significant burden on caregivers [2]. Research indicates that caregivers of PwD experience higher rates of depression compared to the general population [3]–[5]. Neglecting the mental health issues of caregivers negatively impacts their perceived quality of life (QoL) [6]. This reduction in caregivers' QoL diminishes the quality of care provided, subsequently lowering the QoL of PwD [7]. Therefore, it is crucial to identify the risk of depression and caregiver burden and ensure that caregivers receive timely medical assistance.

Despite the acknowledged importance, caregivers often lack the time to seek medical advice for their own conditions due to their care responsibilities and are not screened for mental health issues [8]. Consequently, they overlook symptoms of depression, delaying screening and medical help [7]. While clinically validated screening tools exist [9], they are not well adopted in practice, and more conversational approaches are deemed necessary [10]. Interestingly, social workers, who are often the first point of contact for PwD at day activity centers or other care facilities, regularly hear caregivers discuss their perceived burden and depressive symptoms [11]. This makes social workers an optimal channel to screen for the risk of depression and perceived care burden. Devising accessible screening tools that can be employed in such channels may enable early detection and timely intervention for depression and perceived burden in caregivers.

In this work, we show that the demographic information of caregivers, which social workers can easily access, and casual conversations about caregivers' perceived challenges can estimate the risk of caregivers' depression and burden. More specifically, we propose an analytical pipeline to train models to estimate the risk of depression and perceived burden utilizing 1) caregivers' demographic information only (hereafter denoted as demographic feature set), 2) caregivers' informal narratives about their caregiving experiences only (language feature sets), and 3) both caregivers' demographic information and narratives (combined feature sets). These approaches are based on prior research suggesting that demographic factors like age, race, and socioeconomic status are linked to increased caregiver burden and depression risk [12]. Additionally, studies have shown that narrative language patterns can reveal emotional states and psychological distress [13]. With a specific emphasis on translational impacts to medical practice, using data collected from 65 dementia caregivers, the models trained by the proposed analytical pipeline estimate the risk of depression and care burden based on widely accepted clinical screening measures: the Patient Health Questionnaire-8 (PHQ-8) [9] and the Zarit Burden Interview (ZBI) [14]. Our experiment results demonstrate that our analytical pipeline produces estimation models with acceptable accuracy.

II. METHODS

The study protocol was approved by the Institutional Review Board of Indiana University (#22162).

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TABLE I
GROUPED CHARACTERISTICS OF CAREGIVERS OF PWD (N=65)

Characteristics	Levels	n (%)
Age	18-34 years	59 (90.77)
	35 years and above	6 (9.23)
Sex	Male	48 (73.85)
	Female	17 (26.15)
Race	Black or African American	55 (84.62)
	White	6 (9.23)
	Other	4 (6.15)
Ethnicity	Hispanic or Latino	1 (1.54)
	Not Hispanic or Latino	57 (87.69)
	Unknown or Not reported	7 (10.77)
Location	Urban	43 (66.15)
	Suburban	15 (23.08)
	Rural	7 (10.77)
Employment	Employed full-time	31 (47.69)
	Employed part-time	31 (47.69)
	Other	3 (4.62)
Education	Bachelors or Above	50 (76.92)
	No Bachelors Degree	15 (23.08)
Income	Under \$25,000	12 (18.46)
	\$25,000 - \$50,000	20 (30.77)
	\$50,001 - \$75,000	13 (20.00)
	\$75,001 - \$100,000	17 (26.15)
	Over \$100,000	3 (4.62)
Health Insurance	Yes	48 (73.85)
	No	16 (24.62)
	Not Sure	1 (1.54)
Relationship to loved one	Parent	32 (49.23)
	Other family member	22 (33.85)
	Sibling	6 (9.23)
	Spouse/Partner	7 (10.61)
Years caring	1-2 years	15 (22.73)
	3-5 years	40 (60.61)
	6 years and more	11 (16.67)
Daily caregiving hours	2-4 hours	9 (13.85)
	5-8 hours	36 (55.38)
	More than 9 hours	20 (30.77)
Depressed	No	39 (60.00)
	Yes	26 (40.00)
Burden Level	No High Burden	26 (40.00)
	High Burden	39 (60.00)

A. Study Participants

Recruitment efforts utilized posters in public spaces, churches, daycare centers, and online community support groups (e.g., Reddit) from March through April 2024. The inclusion criteria required participants to be 18 years old or older, have English as their primary language, reside in the United States, and serve as caregivers for PwD. The study included both primary and secondary caregivers, encompassing anyone who assisted patients with daily activities or provided healthcare support (e.g., accompanying patients to appointments and administering medications) [4]. As a result, 65 caregivers of PwD were recruited for the study. TABLE I summarizes the characteristics of our study participants.

B. Data Collection

Data collection involved two main steps: 1) Self-administered surveys including demographic questions, a depression screening component (PHQ-8), and a caregiver burden measure (ZBI); and 2) a semi-structured interview conducted via Zoom. The total scores of the self-administered PHQ-8 and ZBI were computed, which were then used to

estimate the depression-risk group (i.e., PHQ-9 score ≥ 10) and the high burden group (i.e., ZBI score ≥ 20) for target labels. During the Zoom interviews, participants were asked to turn off their video cameras to encourage more candid responses. We employed open-ended interview probes to better understand their symptoms of depression and perceived burden (see Appendix). These probes were developed based on open-ended questions regarding burden and depression, such as in [15]. This approach allowed participants to express themselves freely without feeling pressured by nonverbal cues. The audio data extracted from these recordings were then transcribed and converted into text format for further language analysis using Natural Language Processing (NLP) techniques.

C. Feature Extraction and Selection

1) *Demographic Features*: We used all the characteristics obtained from the demographic survey questions (detailed in TABLE I) as categorical features. Their respective levels served as feature values. To represent these categorical features, we employed one-hot encoding. For missing values in *Race*, *Ethnicity*, and *Income*, we assigned the value “Unknown”. For missing *Location* values, we imputed them using the most frequent value (mode).

For feature selection, we initially excluded features with fewer than five samples per target variable. Next, we employed the Chi-square test of dependence as our feature selection methodology to identify 8 most important features. Previous research by Hua *et al.* [16] suggest that the optimal number of features to avoid overfitting should be limited to $\sqrt{N}$, where N represents the sample size. Hence, we set the maximum number of features to $\sqrt{65} \simeq 8$.

2) *Language Features from Narratives*: To prepare the narrative text data for analysis, we applied several preprocessing steps using spaCy 3.7.5, a Python NLP toolkit [17]. This included converting all text to lowercase, tokenization and punctuation removal.

We used the Bag of Words (BoW) unigram features (single words) frequencies to represent the textual content using three different approaches. First, we computed the word frequencies for all the leveraged unigrams present in the entire narrative (resulting in a total of 2,931 features). Second, we computed the word frequencies of adjectives within the text (resulting in 613 features) because it is reported that in Western languages, adjectives provide more information about actors [18]. Lastly, we computed the word frequencies of the unigrams from every sentence that contained an adjective (resulting in 2,871 features). Here, we hypothesized that these sentences might capture the context surrounding the adjectives, offering more focused insights into caregivers’ experiences. These approaches resulted in three different feature sets, denoted as 1) UNI, 2) ADJ, and 3) ACD hereinafter (Fig. 1.2).

3) *Combined Features*: The selected demographic features were concatenated to the NLP features, each of which was computed as described above, resulting in three different feature sets, denoted as 1) Demo+UNI, 2) Demo+ADJ, and 3) Demo+ACD hereinafter (Fig. 1.2).

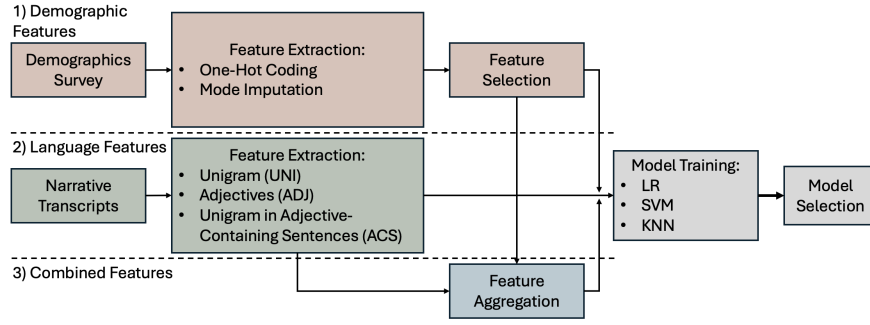
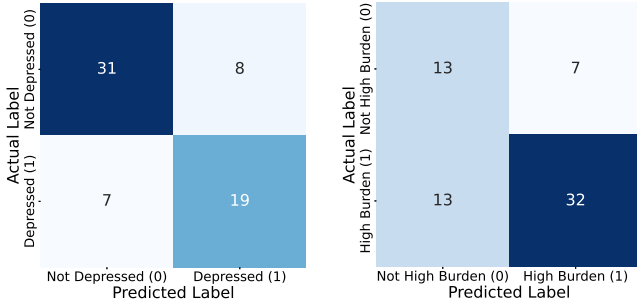


Fig. 1. The diagram describing the analytic pipeline to train the models to estimate the risk of depression and perceived burden, utilizing the three different data sets: 1) demographic, 2) narrative, and 3) combined features.

TABLE II

F1 SCORES FOR THE TRAINED MODELS USING ALL THE COMBINATIONS OF THREE SEVEN FEATURE SETS (I.E., ONE DEMOGRAPHIC FEATURE SET, THREE LANGUAGE FEATURE SETS, AND THREE COMBINED FEATURE SETS) AND THREE DIFFERENT CLASSIFICATION ALGORITHMS

	Depression			High Burden		
	LR	SVM	KNN	LR	SVM	KNN
Demo	0.55	0.49	0.55	0.70	0.76	0.66
UNI	0.57	0.61	0.30	0.70	0.61	0.62
ADJ	0.55	0.54	0.56	0.70	0.69	0.71
ACS	0.69	0.72	0.54	0.66	0.68	0.70
Demo+UNI	0.57	0.62	0.30	0.70	0.61	0.61
Demo+ADJ	0.56	0.54	0.55	0.72	0.73	0.72
Demo+ACS	0.69	0.72	0.54	0.70	0.68	0.70



(a) Best model for depression risk estimation: ACS $\times$ SVM (b) Best model for perceived burden estimation: Demo $\times$ SVM

Fig. 2. Confusion matrices for the best-performing models to estimate the risk of depression and perceived burden, respectively.

D. Model Training and Selection

With the selected features, we trained two estimation models—one for estimating the risk of depression and another for perceived burden—for each data set. To train estimation models, we employed three different classification algorithms, including Logistic Regression (RL), K-Nearest Neighbors (KNN), and Support Vector Machine (SVM) with default hyper-parameters, using scikit-learn 1.2.2. The resulting estimation performance of the trained models were evaluated using F1 scores, all of which were computed via Leave-One-Subject-Out Cross-Validation (LOSO-CV). Among the trained models, the best-performing model for estimating the risk of depression and the perceived burden, respectively, was selected based on F1 scores.

III. RESULTS & DISCUSSION

Twenty-one models were trained using combinations of seven feature sets (i.e., Demo, UNI, ADJ, ACS, Demo+UNI, Demo+ADJ, Demo+ACS) and three classification algorithms (i.e., LR, SVM, KNN) to estimate the risk of depression and the perceived burden, respectively, resulting in a total of 42 models. The F1-scores for all the trained models are reported in TABLE II. Confusion matrices for the best-performing models are presented in Fig. 2.

A. Demographic Features

Demographic features produced the best-performing model to estimate the perceived burden among caregivers, with an F1 score of 0.76 (Demo $\times$ SVM). It is surprising that language features underperformed compared to demographic features. On the other hand, demographic features resulted in poorly performing models for estimating the risk of depression. This suggests that being in a certain situation alone (e.g., a female caregiver providing care for six years or more in a low-income family) can accurately capture the level of care burden. However, not everyone in the same situation experiences the same level of depressive symptoms.

B. Language Features

Language features resulted in the best-performing model when estimating the risk of depression (ACS $\times$ SVM) with an F1 score of 0.72 (TABLE II). These results support our initial hypothesis that sentences containing adjectives may provide more in-depth contextual information regarding caregivers' emotional status, thereby capturing depressive symptoms better. The more detailed results are presented in the confusion matrix (Fig. 2(a)).

C. Combined Features

We initially hypothesized that combined features might be more informative than demographic and language features alone. However, to our surprise, our results suggest that combined features may not add additional nor complementary information. Overall, the F1 scores of the models trained using combined features were almost the same as those of the models trained using language features alone (TABLE II). On a few occasions, combined features (Demo+ACS $\times$ LR and

Demo+ADJ $\times$ SVM) led to greater performance than language features (ACS $\times$ LR and ADJ $\times$ SVM) when estimating high burden (0.70 vs 0.66 and 0.73 vs 0.69, respectively). However, even these features underperformed compared to demographic features (0.76). We conjecture that, since there were a greater number of language features (i.e., 2,871 in ACS) than demographic features (i.e., eight), the additional information added by demographic features was minimal.

IV. CONCLUSIONS

This study investigated the feasibility of estimating the risk of depression and perceived burden among caregivers of PwD by analyzing their demographic information and narratives about caregiving experiences. Our trained models showed promising results with F1 scores of 0.72 for depression and 0.76 for burden estimation. Our findings highlight the potential for early screening using non-clinical data, allowing caregivers to seek timely medical assistance. Future research should focus on expanding the dataset to include a more diverse and larger sample size, which will enhance the generalizability of the findings. Additionally, employing more advanced machine learning techniques, such as large language models, could further improve the accuracy and robustness of the estimations.

APPENDIX

- 1) Think about how you have been feeling in the past few weeks while taking care of your loved one with dementia. Tell us about any recent moments that stand out to you, whether that being positive or challenging.
- 2) Think about how caregiving has affected your overall well-being in the past few weeks, particularly regarding your relationships with family, friends, or colleagues. Tell us about any changes or shifts that you have noticed in these relationships.
- 3) Over the past few weeks, have you, as a caregiver, noticed any changes in your decision-making process? If yes, please elaborate.
- 4) Think about your self-care practices in the past few weeks to manage the daily stresses of caregiving. Tell us your strategies that have been particularly helpful in coping with the daily stresses of caregiving.
- 5) In the past few weeks, have any factors such as professional commitments, financial constraints, or health changes significantly influenced how you provide care? If yes, please elaborate on these factors and how they have impacted your caregiving responsibilities.
- 6) As a caregiver, do you sometimes feel concerned about what the future holds for you in that role? If so, could you share more about it? What kind of help or support would be helpful to you as a caregiver?

ACKNOWLEDGMENT

The authors extend their gratitude to Prof. Seungyeon Lee from the University of Central Arkansas for reviewing the interview probes and providing valuable suggestions.

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